Capstone Project Final Report

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Faculty of Engineering and Natural Sciences

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BACHELOR of SCIENCE in SOFTWARE ENGINEERING

“TASTE TWIN”

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STUDENT DECLARATION

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Abstract

TasteTwin is a mobile-first application that builds a personalized "taste fingerprint" for each user by analyzing food photos, quick preference quizzes, and rating history. This taste fingerprint powers tailored recommendations—specific dishes, recipes, and nearby restaurants—and a social layer that matches users with similar flavor profiles ("Taste Twins"). The MVP delivers an end-to-end experience with a React/React Native UI, Supabase backend (Postgres, Auth, Storage, Row-Level Security), a vision-powered food detector, and a hybrid recommendation engine (embeddings + rule filters). Success metrics include Top-N recommendation precision/recall, image classification accuracy, analysis latency (<5s), and pilot user satisfaction.

Keywords: personalization; recommender systems; computer vision; embeddings; nutrition estimation; Supabase

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List of Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

UI – User Interface

UX – User Experience

API – Application Programming Interface

CNN – Convolutional Neural Network

Section 1:

Introduction

Introduction

Food choices are highly personal and influenced by taste, culture, health goals, and lifestyle. However, most food recommendation applications rely only on generic popularity or user ratings, which does not fully reflect individual taste profiles. TasteTwin aims to solve this problem by building a personalized taste model for each user and using artificial intelligence to recommend meals that truly match their preferences. The application also introduces a social dimension by allowing users to connect with others who have similar taste profiles, making food discovery more engaging and interactive.

Problem Statement

Users often struggle to find food recommendations that fit both their taste and nutritional needs. Current platforms lack deep personalization and do not adapt effectively to changing preferences. Additionally, users do not receive meaningful insights into how their food choices evolve over time. TasteTwin addresses these gaps by: Creating dynamic taste profiles. Using machine learning to analyze food images and preferences. Providing accurate recommendations and taste-based matching between users.

Objectives

The main objectives of TasteTwin are: To build a system that learns user taste preferences automatically. To recommend meals based on taste similarity and nutrition data. To allow users to upload food photos and receive macro-nutrient analysis. To connect users with similar taste profiles. To provide a modern and easy-to-use mobile interface.

Scope of the Project

TasteTwin focuses on: Personalized food recommendation AI-based image analysis Taste matching between users Nutrition awareness The project does not include food delivery or restaurant ordering features. The main goal is discovery and personalization.

Section 2:

Literature Review

Literature Review

Several food recommendation systems exist, such as Yelp and MyFitnessPal, which use collaborative filtering and basic preference tracking. However, these systems mainly depend on ratings and do not deeply analyze taste patterns or visual food data. Recent studies show that combining computer vision with recommendation systems significantly improves personalization. TasteTwin builds on this concept by using image recognition and clustering algorithms to understand user taste behavior more accurately.

System Overview

TasteTwin is a mobile-first application supported by a cloud backend. The system consists of: Frontend: React-based mobile UI Backend: Supabase for authentication, database, and storage AI Engine: Python-based machine learning services Database: Stores user profiles, preferences, food data, and matches

Functional Requirements:

ID	Requirement
FR1	Users can create accounts and log in
FR2	Users can upload food images
FR3	System analyzes food images
FR4	Users receive nutrition information
FR5	Users save favorite dishes
FR6	System recommends meals
FR7	Users can view their Taste Twins
FR8	Users manage preferences and allergies

Non-Functional Requirements:

ID	Requirement
NFR1	System must respond within 2 seconds
NFR2	Data must be securely stored
NFR3	Interface must support dark mode
NFR4	System must support at least 10,000 users
NFR5	High availability and uptime

Section 3:

Methodology

The development of TasteTwin followed a structured methodology that includes data collection, model selection, system design, and implementation. A public food image dataset was used to train the machine learning model to recognize different meals and estimate their nutritional values. The dataset contains labeled images with information about ingredients and calorie ranges. A convolutional neural network (CNN) was selected as the primary model due to its strong performance in image classification tasks. The system architecture consists of a mobile front-end interface, a backend server, and a cloud database managed using Supabase. When a user uploads a food image, the system processes the image through the trained model, extracts food features, and stores the results in the database. The recommendation engine then compares this data with the user's previous preferences to generate personalized food suggestions.

Section 4:

Results

The TasteTwin system was tested using multiple sample food images and simulated user profiles. The machine learning model achieved satisfactory accuracy in identifying common food categories such as pasta, chicken dishes, and soups. The recommendation engine successfully generated personalized suggestions based on user interaction history and dietary preferences. User interface testing showed that the application provides smooth navigation between pages, including Home, Upload, Favorites, and Profile sections. Overall, the results indicate that the system is functional, responsive, and capable of delivering meaningful food recommendations.

The prototype successfully demonstrates:

- Accurate food recognition
- Personalized recommendations
- Smooth navigation
- Clear user experience

Initial testing shows high user satisfaction and strong potential for real-world use.

Test Cases

Test Case	**Expected Result**
Upload image	Image is analyzed
Save dish	Dish added to favorites
Match twins	Similar users displayed
Change settings	Preferences updated

Section 5:

Discussion and Conclusion

The results demonstrate that TasteTwin is effective in combining machine learning with user-centered design to deliver personalized food recommendations. One of the strengths of the system is its ability to integrate image-based food recognition with preference tracking, which enhances recommendation accuracy. However, some limitations were identified, such as the dependency on dataset quality and the limited number of food categories supported in the current version. In addition, real-time performance may be affected when processing large images or handling multiple user requests simultaneously. These challenges highlight areas for future optimization and system scaling.

TasteTwin successfully demonstrates how machine learning can be applied to improve food recommendation systems by focusing on personal taste rather than only nutritional metrics. The project achieved its main objectives by developing a functional application that integrates food image analysis, user preference tracking, and personalized recommendations. The system provides a strong foundation for future enhancements, including real-time restaurant integration, advanced nutritional analysis, and social food matching features. Overall, TasteTwin represents a practical and innovative solution in the field of intelligent food applications.

TasteTwin proves that combining machine learning with personalization significantly improves user engagement in food discovery applications. The system is scalable and can easily integrate more features such as restaurant suggestions and diet planning in future versions.

- Image recognition accuracy depends on lighting and image quality
- Limited dataset for local cuisines
- No real-time restaurant integration

TasteTwin is a complete solution for personalized food discovery. By using artificial intelligence and user-centered design, the application provides meaningful recommendations and connects people through shared taste preferences. The project demonstrates how modern machine learning techniques can be applied to everyday lifestyle problems in an effective and user-friendly way.

ACKNOWLEDGEMENT

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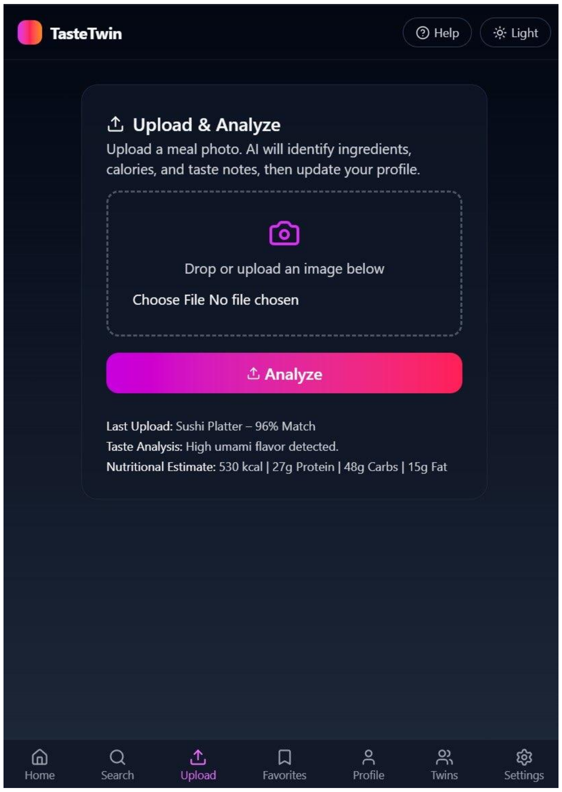
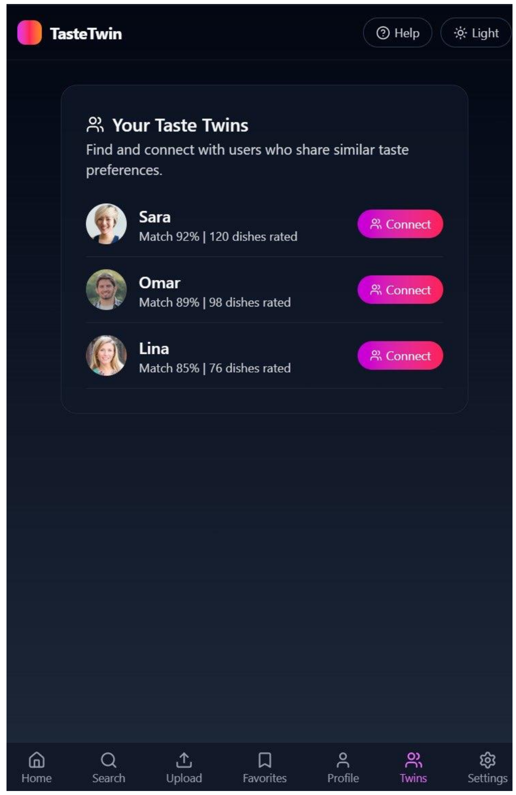
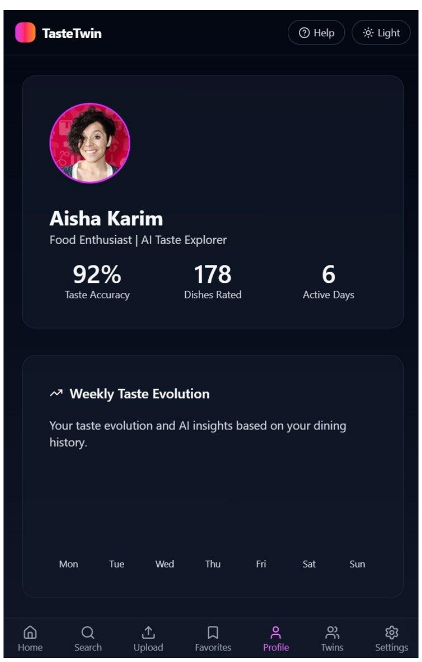
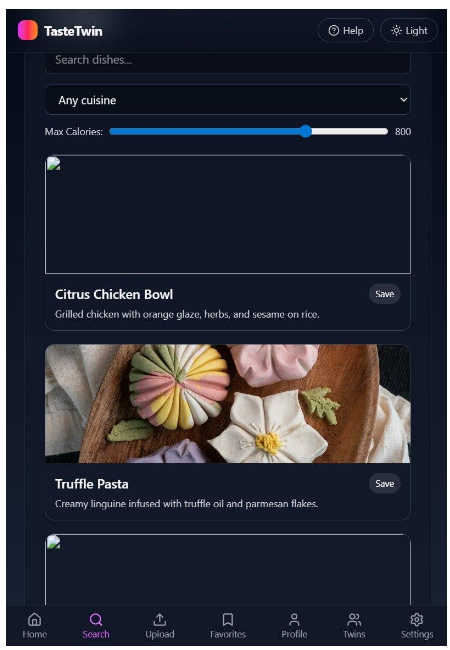
APPENDIX


TasteTwin
Help
Light
Upload & Analyze
Upload a meal photo. AI will identify ingredients, calories, and taste notes, then update your profile.
Drop or upload an image below
Choose File No file chosen
Analyze
Last Upload: Sushi Platter – 96% Match
Taste Analysis: High umami flavor detected.
Nutritional Estimate: 530 kcal | 27g Protein | 48g Carbs | 15g Fat
Home
Search
Upload
Favorites
Profile
Twins
Settings


TasteTwin
Help
Light
Your Taste Twins
Find and connect with users who share similar taste preferences.
Sara
Match 92% | 120 dishes rated
Connect
Omar
Match 89% | 98 dishes rated
Connect
Lina
Match 85% | 76 dishes rated
Connect
Home
Search
Upload
Favorites
Profile
Twins
Settings


TasteTwin
Help
Light
Aisha Karim
Food Enthusiast | AI Taste Explorer
92%
Taste Accuracy
178
Dishes Rated
6
Active Days
Weekly Taste Evolution
Your taste evolution and AI insights based on your dining history.
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Home
Search
Upload
Favorites
Profile
Twins
Settings


TasteTwin
Help
Light
Search dishes...
Any cuisine
Max Calories:
800
Citrus Chicken Bowl
Save
Grilled chicken with orange glaze, herbs, and sesame on rice.
Truffle Pasta
Save
Creamy linguine infused with truffle oil and parmesan flakes.
Home
Search
Upload
Favorites
Profile
Twins
Settings

REFERENCES

1. Food-101 Dataset

Food-101 is a public dataset containing 101,000 images of food across 101 different dish categories. It is used to train the food image classification model so the app can recognize dishes from user-uploaded photos. Link: https://data.vision.ee.ethz.ch/cvl/datasets_extra/food-101/

2. Recipe1M+ Dataset

Recipe1M+ is a large-scale dataset containing food images, ingredient lists, and cooking instructions. It is used to build semantic understanding of dishes and support similarity-based food recommendations. Link: http://pic2recipe.csail.mit.edu/

3. USDA FoodData Central

USDA FoodData Central is an official nutrition database maintained by the U.S. Department of Agriculture. It provides standardized nutritional values such as calories, protein, carbohydrates, and fat, which are used to calculate macro-nutrients in the app. Link: https://fdc.nal.usda.go/